A language network in the individualized functional connectomes of over 1,000 human brains doing arbitrary tasks

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A century and a half of neuroscience has yielded many divergent theories of the neurobiology of language.[1–10] Two factors that likely contribute to this situation include (a) conceptual disagreement about language and its component processes, and (b) intrinsic inter-individual variability in the topography of language areas. Recent functional magnetic resonance imaging (fMRI) studies of small numbers of intensively scanned individuals have argued that a language-selective brain network emerges from correlations (individualized functional connectomics, iFC) in task-free (e.g., rest) or task-regressed activation timecourses.[11–13] We tested this hypothesis at scale and evaluated its practical utility for task-agnostic language localization: we applied iFC to the entire history of scanning in the Fedorenko lab: 1,957 (fMRI) scanning sessions (1,199 unique brains), each consisting of diverse tasks.

Methods: We developed a data-driven network parcellation procedure (Fig 1) for arbitrary fMRI data from diverse task and naturalistic paradigms. In brief, within each scanning session, irrespective of task, we used functional connectivity (correlation in low-frequency activity) between brain regions to compute probabilistic assignments of voxels to putative networks. We then assigned interpretable labels to the networks based on topographic similarity to reference network atlases defined by prior work, including a language network.[12,14] Finally, we studied the functional properties of these networks by comparing their topography to diverse task contrasts within the same individuals across domains, including language, theory of mind, executive functions, numerical cognition, music perception, and high-level visual processing. To ensure data independence, a critical language localizer task contrasting sentences with nonword lists (S-N)[15] was held out from parcellation.

Results: We found that our iFC-derived language network (LangFC) replicated prior task-based claims about the existence and functional properties of an integrated language network.[9] LangFC was similar in topography to both the reference atlas for the language network (Fig 5A) and the individual's task-evoked response to language (Fig 5B). LangFC was highly selective for language, with strong responses to linguistic tasks and weak/negative responses to nonlinguistic tasks (e.g., hard vs. easy spatial working memory, Fig 2C). LangFC dissociated spatially (Fig 5A) and functionally (Fig 2) from nearby networks (e.g., frontoparietal network A, default network B, and the auditory network) and was more stable within participants across sessions than between participants (Fig 3A), indicating reliable inter-individual variability in structure-function tethering for language. Further analyses showed that LangFC was robust to task state (Fig 5D), identifiable from only a few minutes of fMRI data (Fig 3B), and successful as a drop-in replacement for task-based localization in reanalyses of a prior experiment (Fig 4).[16]

Conclusion: We found that iFC reveals a left-lateralized frontotemporal network that is more stable within individuals than between them, robust to task state, and selective for language. These results support the hypothesis that this network is a key structure in the functional organization of the adult brain and show that it can be recovered retrospectively from arbitrary imaging data even without localizer tasks, with implications for neuroscience, neurosurgery, and neural engineering.

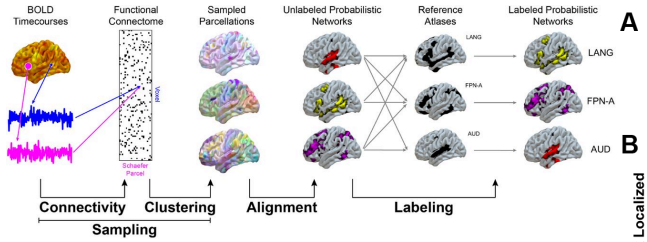

Fig 1: Parcellation method. Correlations between voxelwise time series are discretized and then repeatedly cluster (sampling), then aligned into a common space and labeled by closest fit to a set of *a priori* networks.

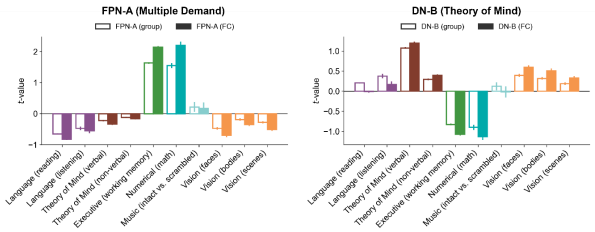

Fig 2: Different precision-localized brain networks show different functional tuning from LangFC, in line with prior expectations of function in these networks.

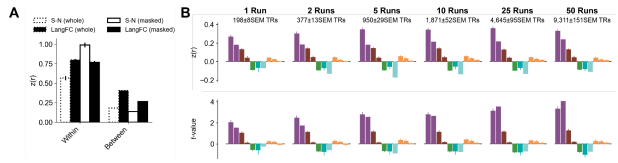

Fig 3: A. Both LangFC and task maps for sentences over nonwords (S-N) show higher Fisher-transformed spatial correlation z(r) with participants across sessions than between participants, where whole brain or masked to a priori language parcels. **B.** LangFC shows language selectivity even when estimated from only one run (for color labels, see Fig 2), but increasing in selectivity (*t*-value) with more data (50 runs).

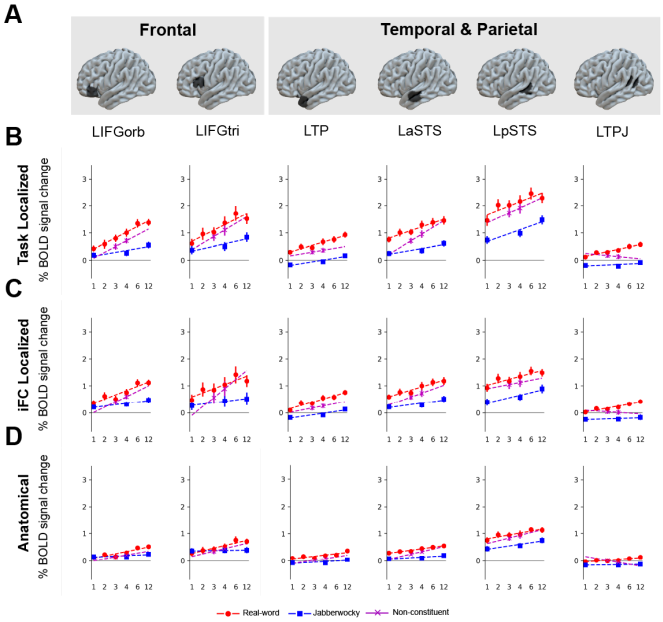

Fig 4: iFC (**C**) largely recovers the sensitivity of task-based localization (**B**) over group-anatomical approaches (**D**) in a prior study's [16] parametric manipulation of lexicality (real vs Jabberwocky) and structural complexity (constituent length, *x*-axis).

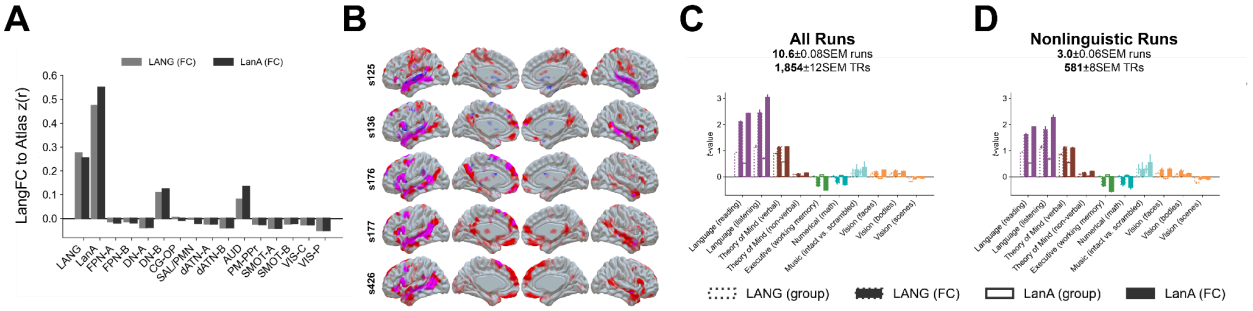

Fig 5: A. Fisher-transformed spatial correlation z(r) of LangFC as labeled relative to a task-based atlas (LanA) vs. a connectivity-based atlas (LANG) in comparison to 16 reference atlases for *a priori* brain networks. **B.** A sample of 5 individuals in the study comparing the distribution of LangFC (blue) to a sentences vs nonwords task contrast (red), with overlap in magenta. **C.** Average *t*-value across diverse tasks within LangFC as labeled relative to LanA (solid outlines) and LANG (dotted outlines) for precision-localized (filled) vs. group-anatomical (unfilled) networks.

References: [1] Geschwind 1970; [2] Hickok & Poeppel 2007; [3] Hagoort 2005; [4] Friederici 2017; [5] Bornkessel-Schlesewsky & Schlesewsky 2013; [6] Matchin & Hickok 2020; [7] Pallier et al. 2011; [8] Brennan et al 2012; [9] Fedorenko et al. 2024, *Nat Rev Neuro*; [10] Aliko et al. 2023; [11] Braga et al. 2020; [12] Du et al. 2024, *J Neurophys*; [13] Salvo et al. 2024; [14] Lipkin et al. 2022; [15] Fedorenko et al. 2010; [16] Shain et al. 2024, *J Cog Neuro*